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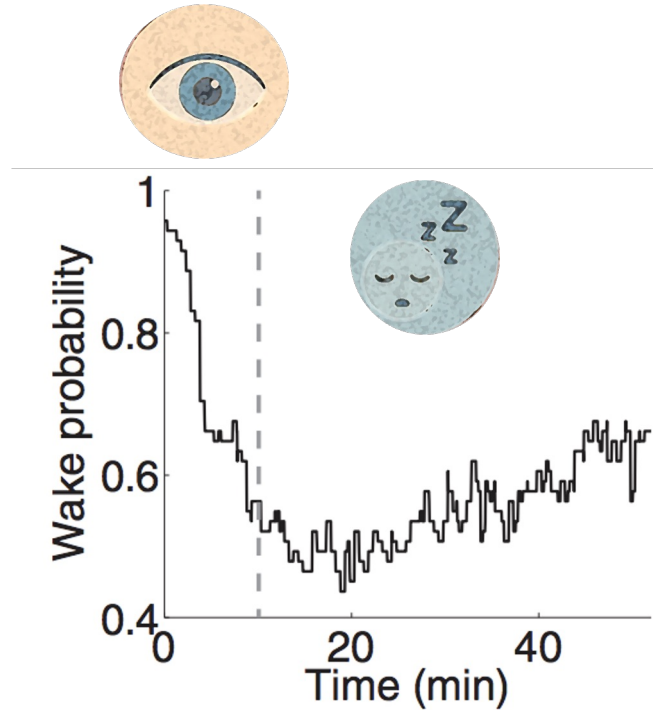
Dynamic brain-body states: implications for reproducible and interpretable neuroimaging

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Neuroimaging and Brain Dynamics Lab

Dept. of Electrical and Computer Engineering
Vanderbilt University

12-05-2025

Variations in attention & vigilance are ubiquitous (and occur during fMRI)

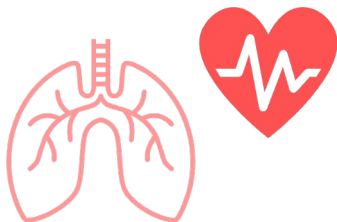


Dynamic neuromodulatory states

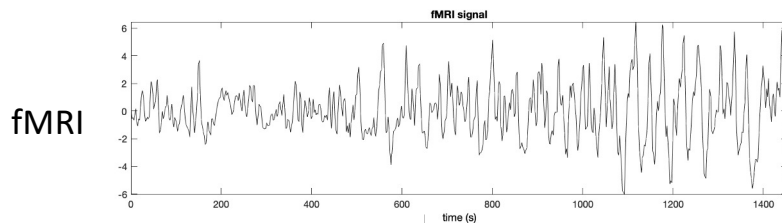
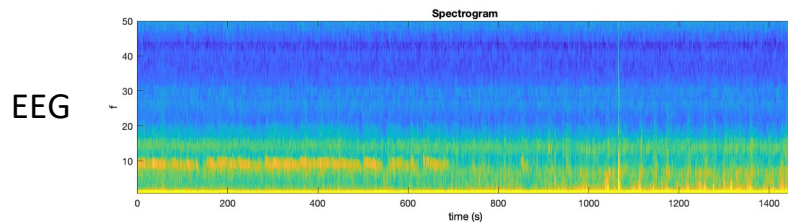
Behavior, cognition



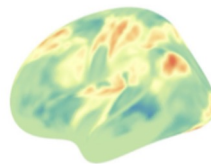
Physiology, autonomic activity



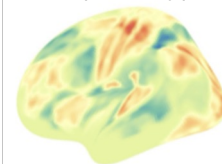
Brain activity



(alert)



(drowsy)



C. Li et al., 2025

Importance of considering dynamic states in fMRI

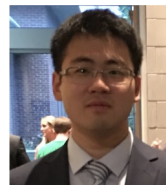
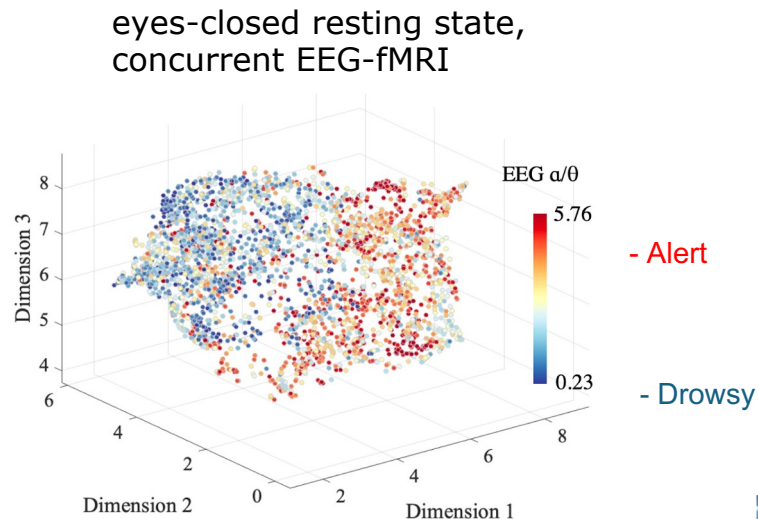
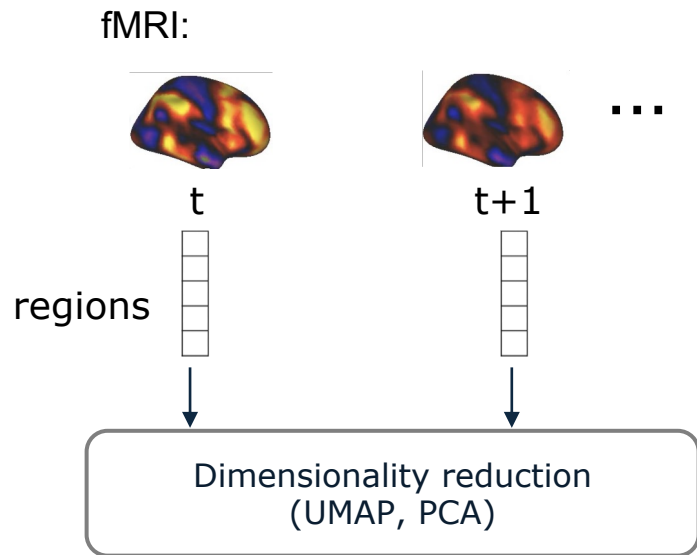
- Source of variability within/between scans
 - Understand & model it to obtain more accurate, sensitive inferences
- Group differences (e.g., patients v. controls)
 - Sleep quality, medication, anxiety
- Crucial aspect of human physiology and function!
 - Fundamental neuroscience
 - Clinical biomarkers
 - I.e., not just a “confound”

→ **Considering dynamic, brain-body states can enable deeper and more powerful neuroimaging studies**

Investigating signatures of vigilance in fMRI

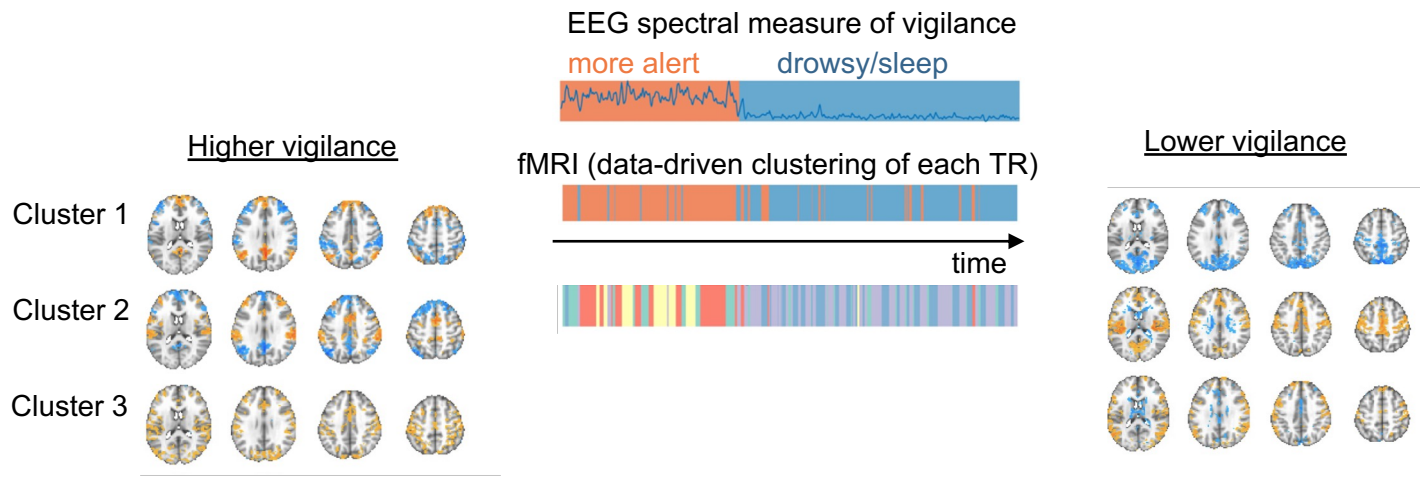
- How are changes in vigilance reflected in whole-brain fMRI patterns?
- Can we detect variations in vigilance from fMRI

Do time-resolved fMRI patterns encode states of alertness?



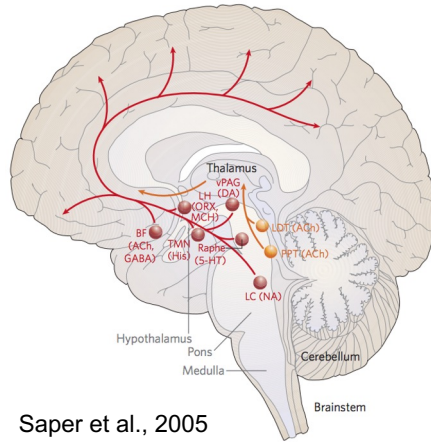
Alignment of low-dimensional fMRI patterns with alertness

Resting-state scan from 1 subject:

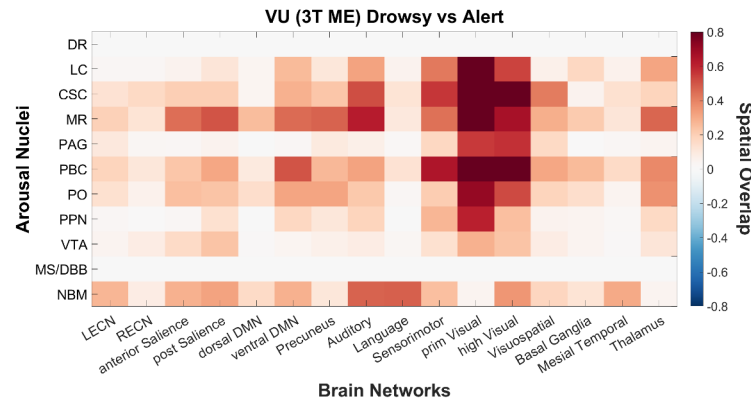
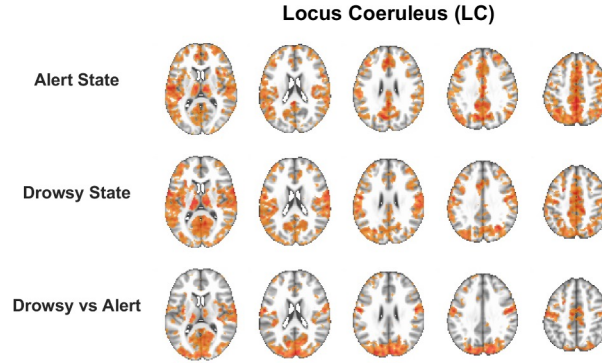


- Time-resolved patterns in fMRI data are linked with vigilance
- Implications for interpretation of inter- and intra-individual spontaneous activity

Vigilance state-dependent subcortical connectivity



Saper et al., 2005

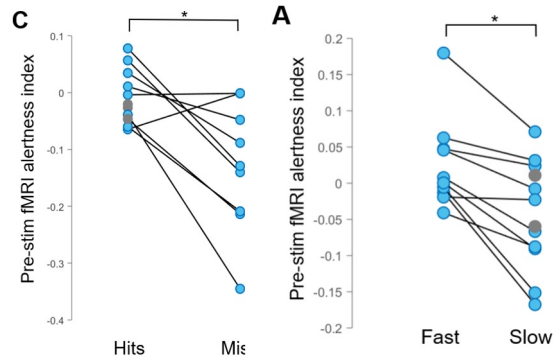
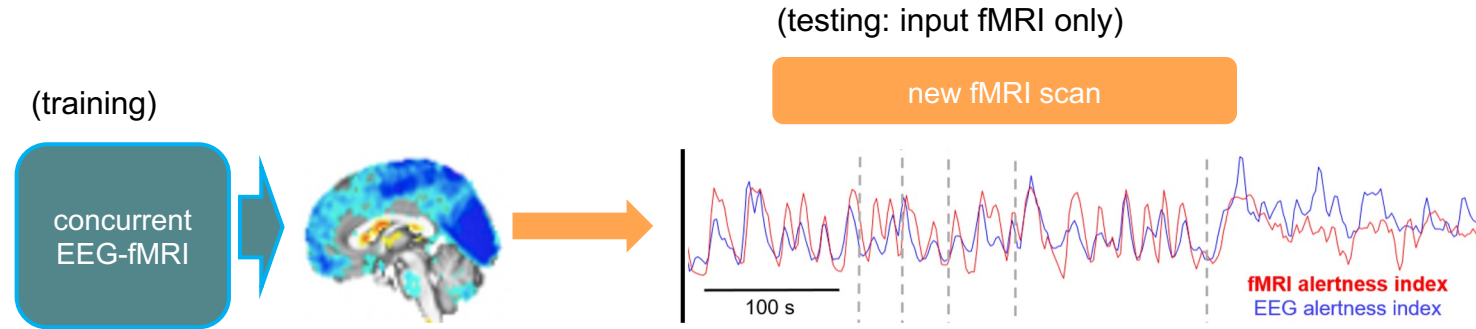


H. Pourmotabbed et al.,
Imaging Neuroscience 2025

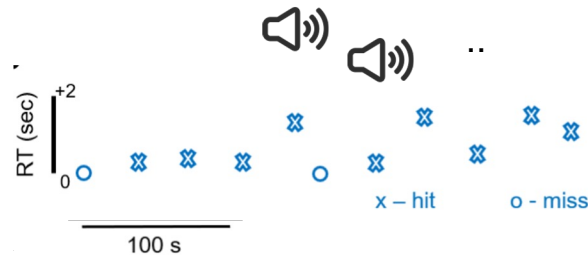
Investigating signatures of vigilance in fMRI

- How are changes in vigilance reflected in whole-brain fMRI patterns?
- **Can we detect variations in vigilance from fMRI**
 - Most fMRI studies don't monitor vigilance, and may not have behavioral measures (e.g., resting state)
 - fMRI-based vigilance readout would enable state-informed analysis

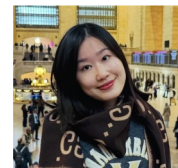
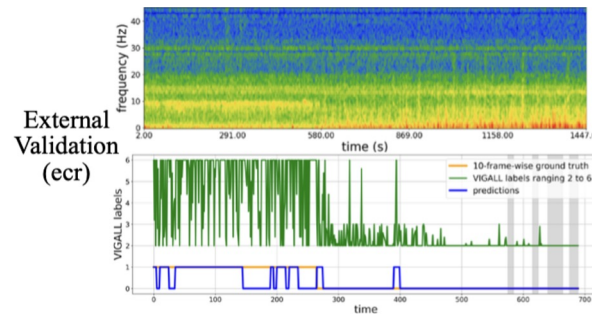
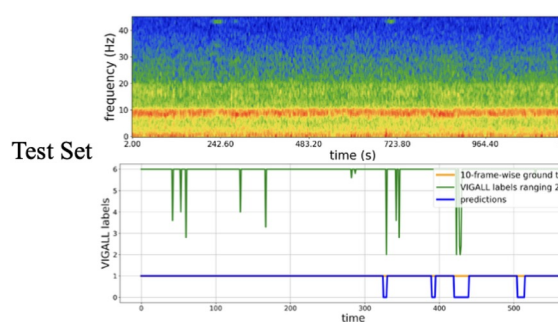
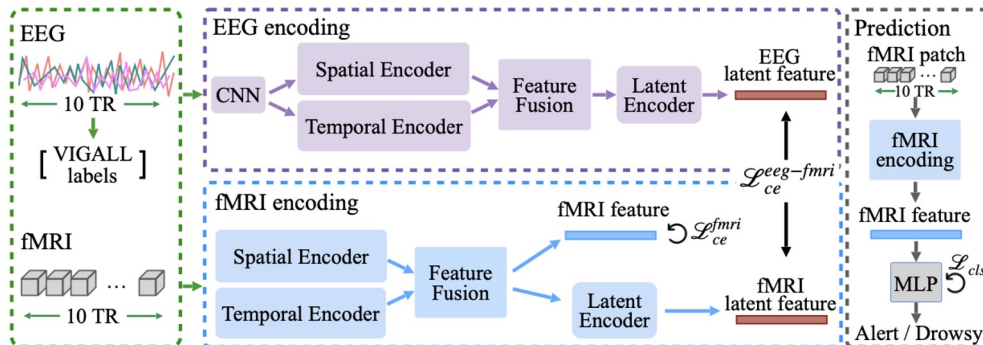
Decoding alertness levels from fMRI



predicting upcoming behavior

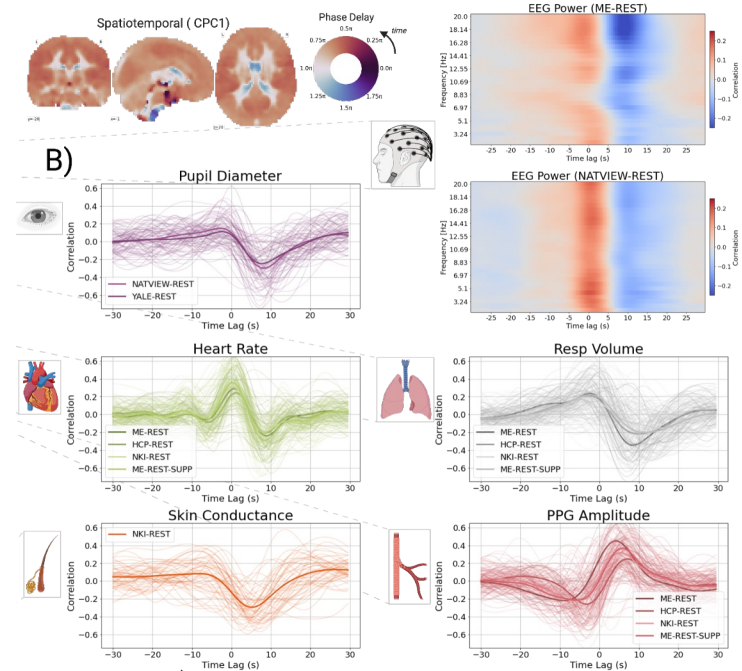
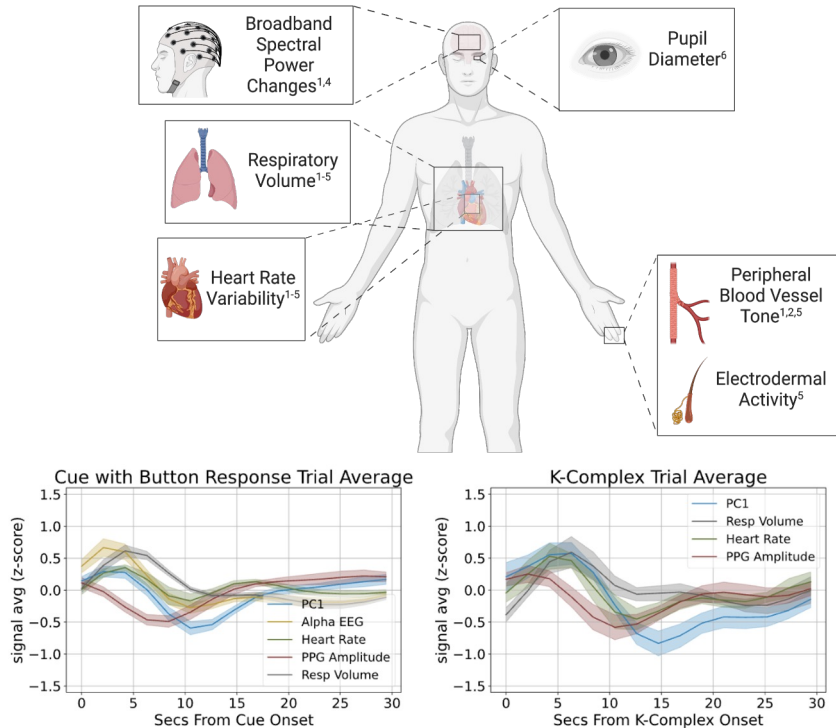


Decoding alertness from brain activity: cross-modal learning framework



Brain-body interactions: Integrating imaging and peripheral physiology

Large-scale mode of coordinated brain-body physiology

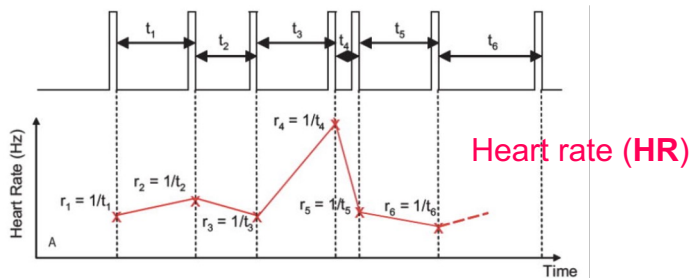
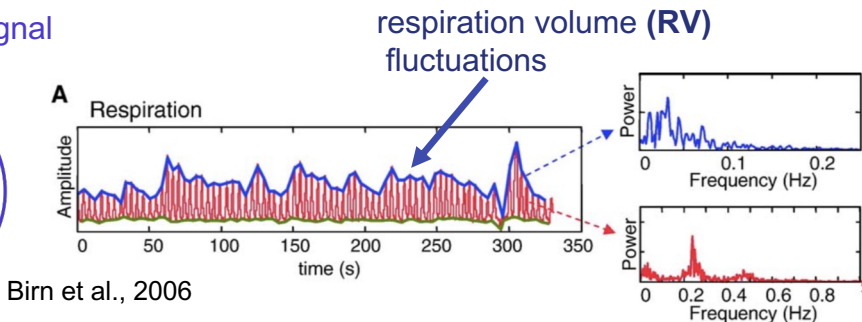
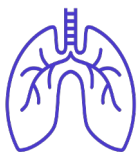


Taylor Bolt et al., *Nat. Neurosci.* 2025
(see also: Gu et al., 2022; Raut et al., 2021; Yang et al., 2025)

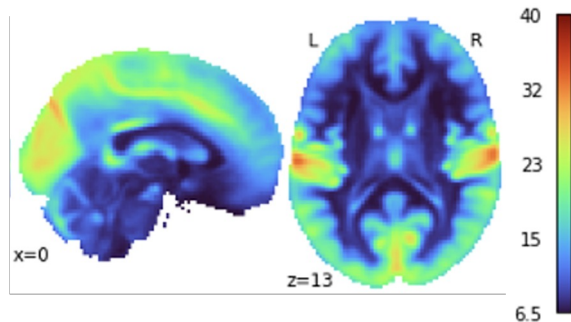
Relating fMRI & peripheral physiology - for reproducibility & neuroscience

Physiological effects in fMRI

respiration belt signal

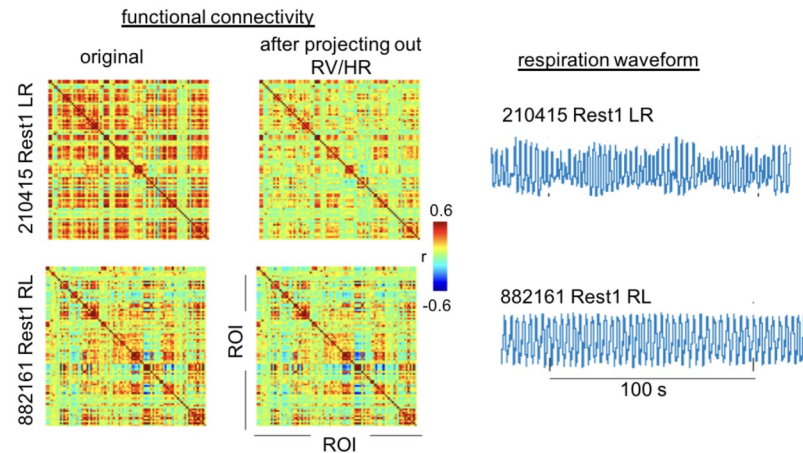
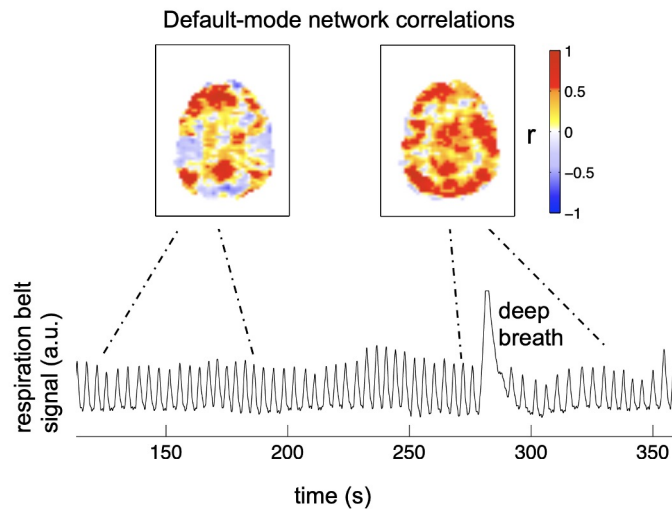


% temporal variance explained in fMRI by RV + HR



(average over 375 subjects, HCP)

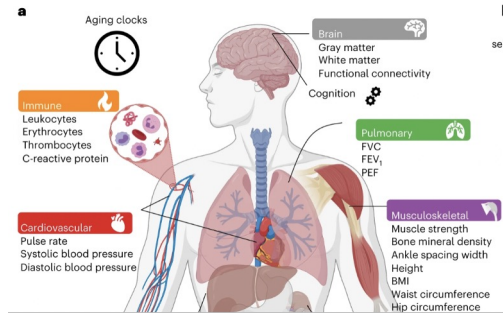
Physiological effects in fMRI as “noise”



Chen, Rubinov, & Chang, 2017

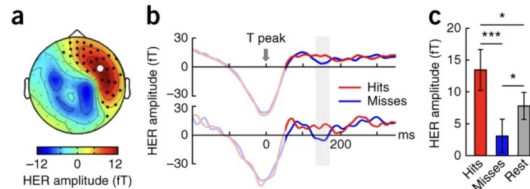
Physiological effects in fMRI as “signal”

Multi-organ aging network



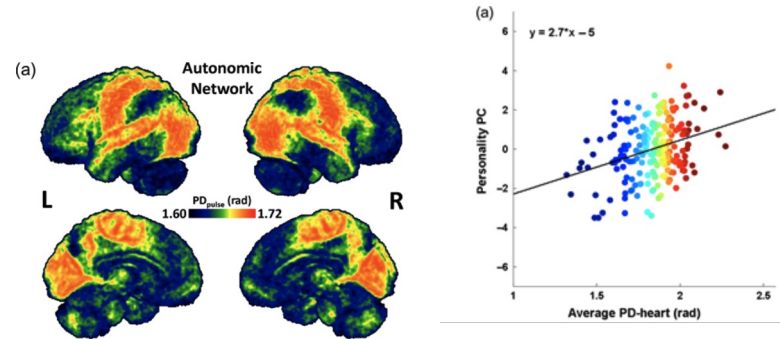
Tian et al., 2023

Figure 3: Neural events locked to heartbeats before stimulus onset predicted conscious perception of the stimulus.



HD Park et al., 2014

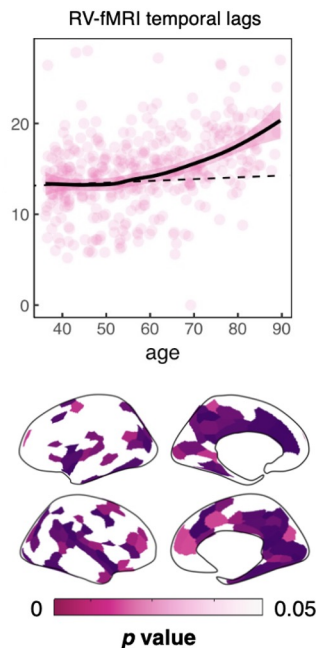
BOLD-autonomic synchrony relates to emotion and personality measures



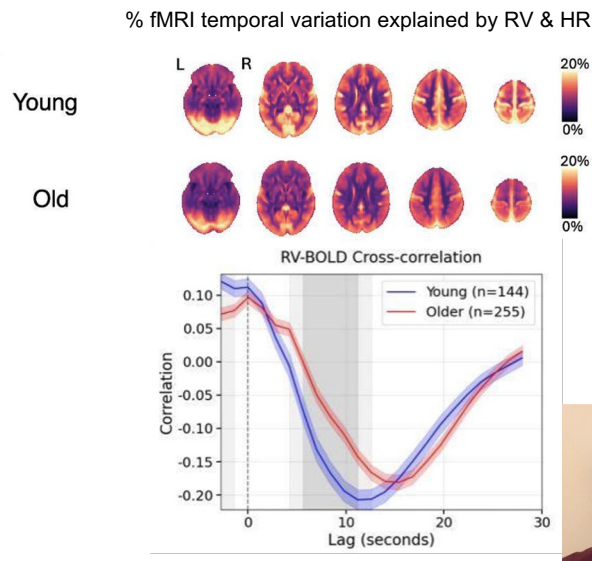
Shokri-Kojori et al., 2018

Physiological effects in fMRI as “signal”

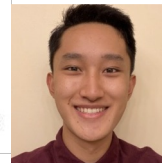
Physiological dynamics in aging



Fan et al., *Comms. Biol.*, 2025
(collab. with Jingyuan Chen et al.)

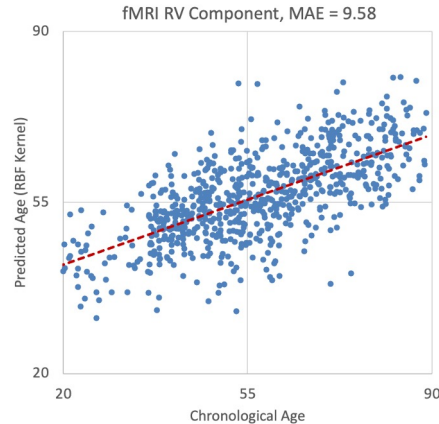


R. W. Song et. al., *Imaging Neuroscience*, 2025

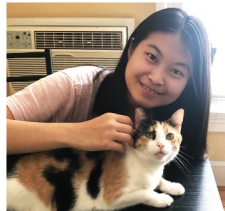


Physiological effects in fMRI as “signal”

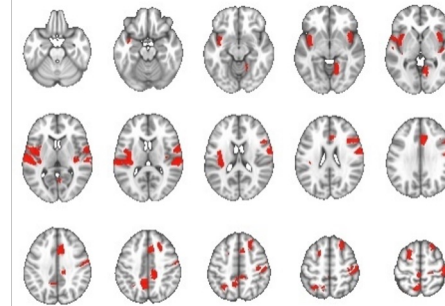
Physiological patterns in aging



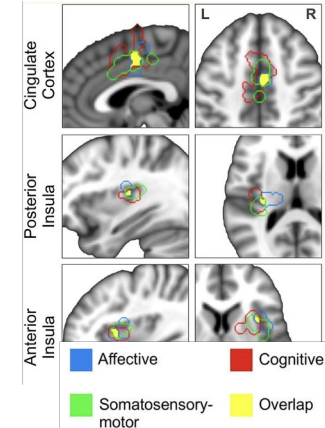
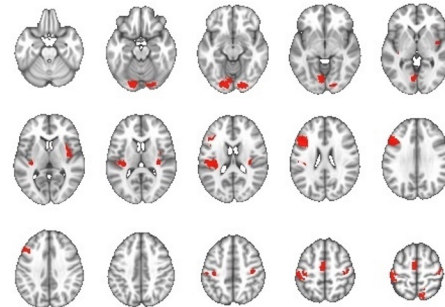
Shiyu Wang et al.,
in revision



RV top 20% features - SHAP importance



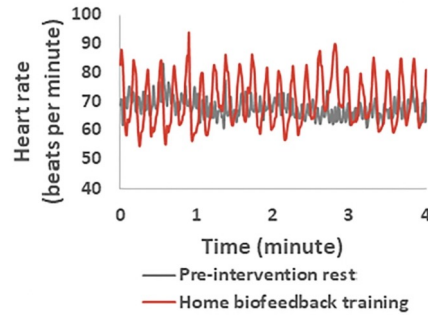
HR



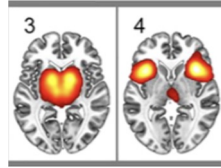
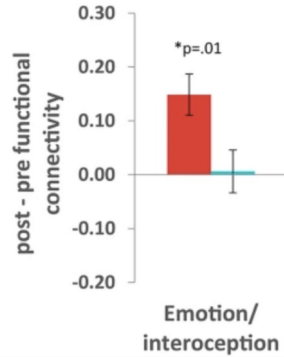
Beissner et al., 2013 – conjunction
analysis of autonomic modulatory
regions in 3 task categories

Modulating brain networks via physiologic interventions

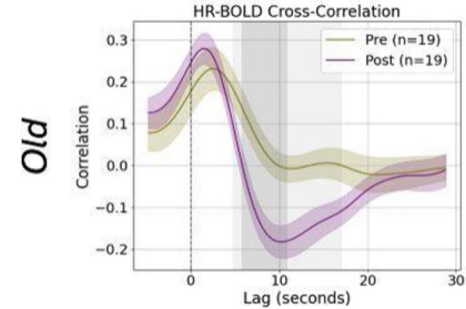
Heart rate variability biofeedback training
alters brain network connectivity



Nashiro et al., 2022



Heart rate - fMRI correlations:
pre/post HRV biofeedback



Song et al., 2025

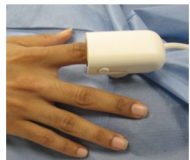
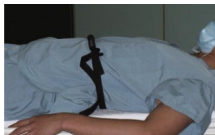


Collaborations
w/ Mara Mather Lab, USC

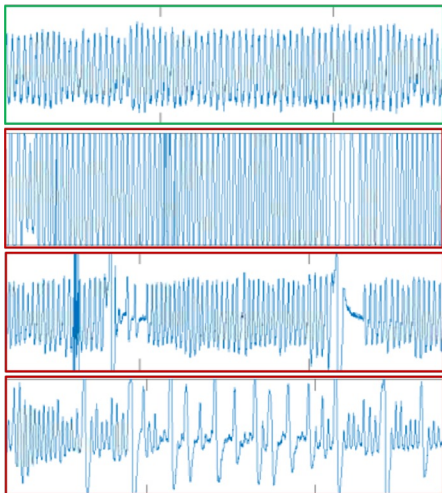
Tools & methods for investigating concurrent brain-body signals

Barriers to studying physiological effects in fMRI

breathing belt



photoplethysmogram (PPG)



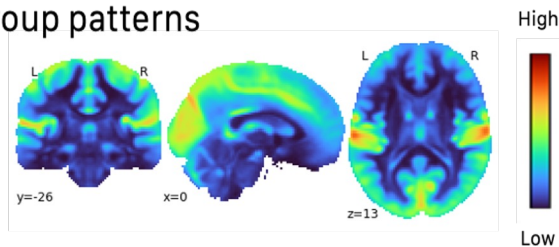
- Monitoring peripheral physiology during fMRI allows investigating brain-body physiology
- Peripheral physio signals are often absent or corrupted
- Time and effort needed to quality-check and correct artifacts
- **Can we infer physiology directly from fMRI?**
see also: Beall et al., 2007; Ash et al. 2010; Tong et al., 2012; Aslan et al. 2019, Agrawal et al., 2020
- **Can we facilitate rapid QA and visualization of physio signals?**

Inferring physiology from fMRI:

We focus on *low-frequency respiration volume (RV)* & *heart rate (HR)*

Influence of Respiration and Heart Rate on fMRI

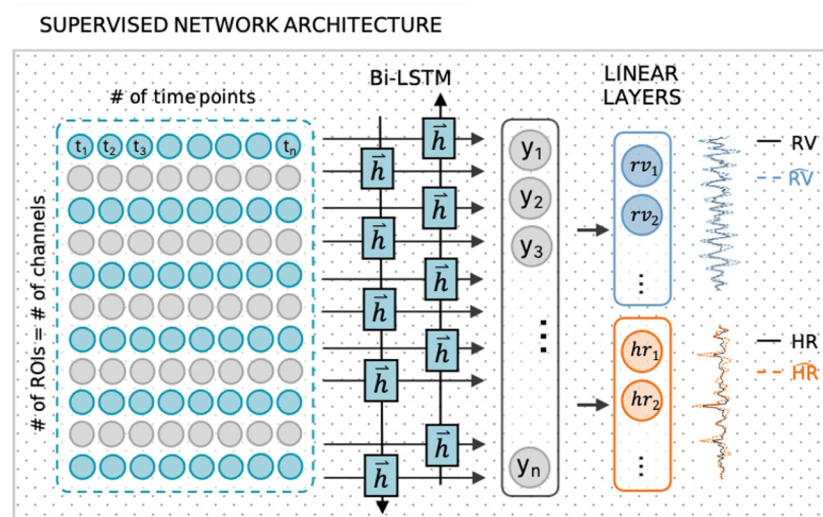
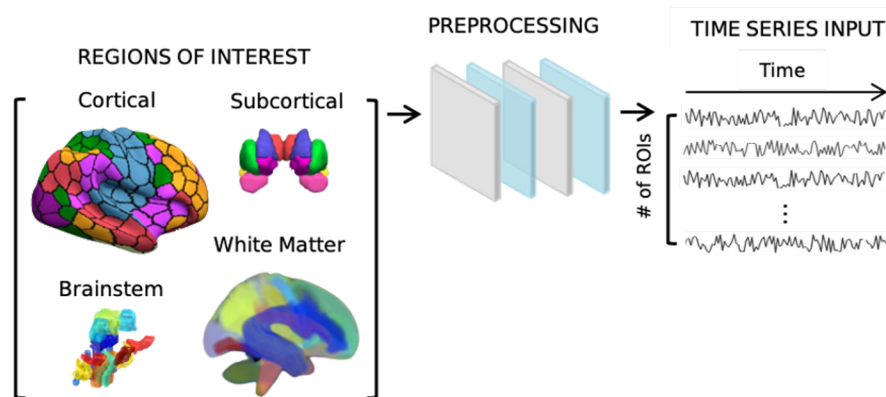
Group patterns



Bayrak et al. 2025

- Physiological recordings can tell us about the state of an individual during a scan (Gray et al., 2009)
- Gastric resting-state network (Rebollo et al., 2018)
- Emotions, personality, and cognitive effort could be linked to the interactions between brain regions and autonomic activity (Shokri-Kojori et al., 2018)
- Heart rate variability biofeedback training alters brain connectivity (Jung et al. 2024, Nashiro et al., 2022)

DeepPhysioRecon: Inferring low-frequency RV and HR dynamics directly from fMRI



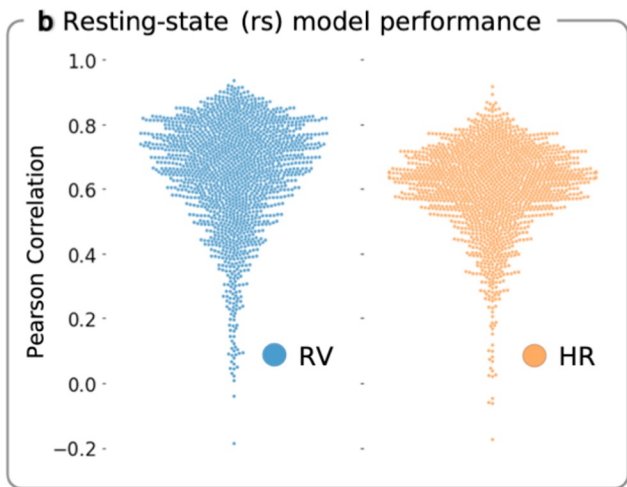
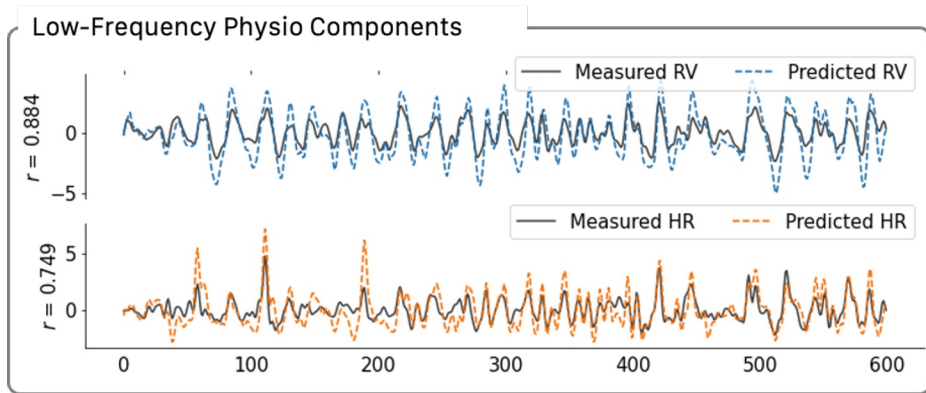
Bayrak et al., 2025 Imaging Neuroscience

Salas et al. 2021 Neuroimage

Bayrak et al. 2020 and 2021 MICCAI

Training and evaluation (HCP Young Adults: 22-35 y.o.)

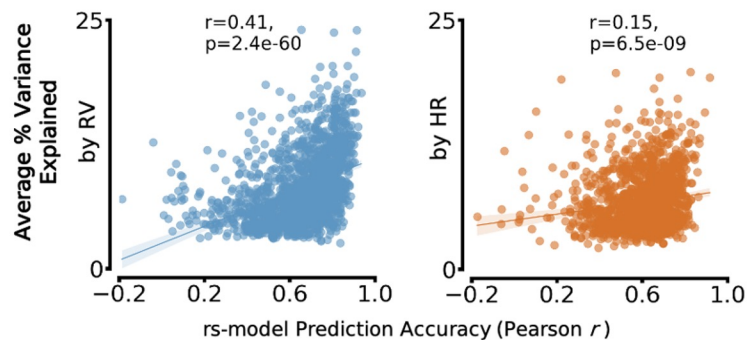
Resting-state data, 5-fold cross-validation



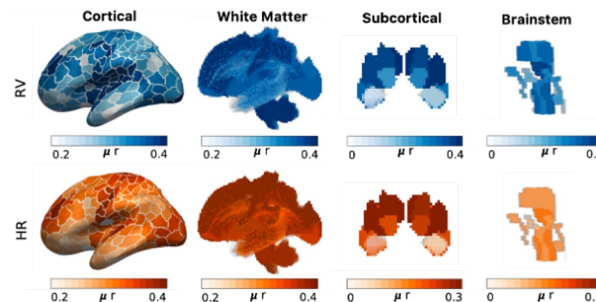
1500 scans from 375 subjects median $r \sim .689$ for RV and $r \sim .627$ for HR

Factors driving model performance

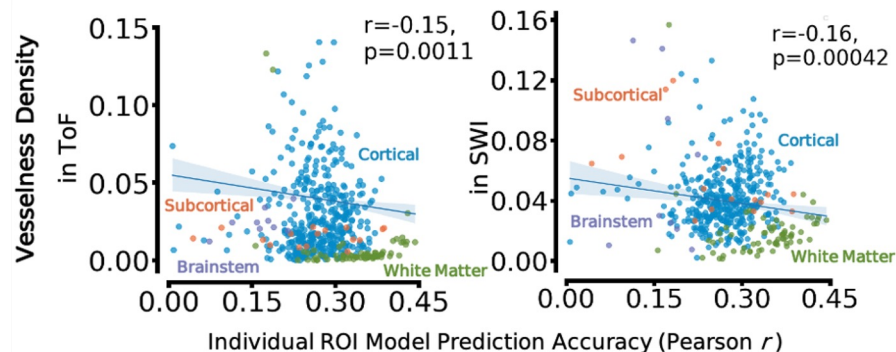
Percent variance



Regional Predictiveness

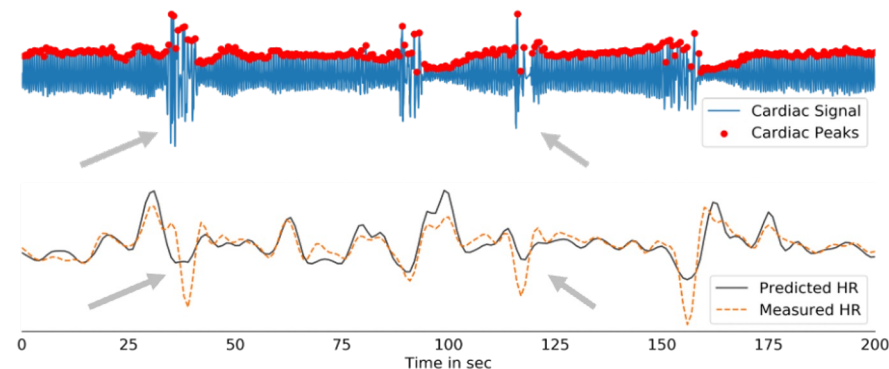
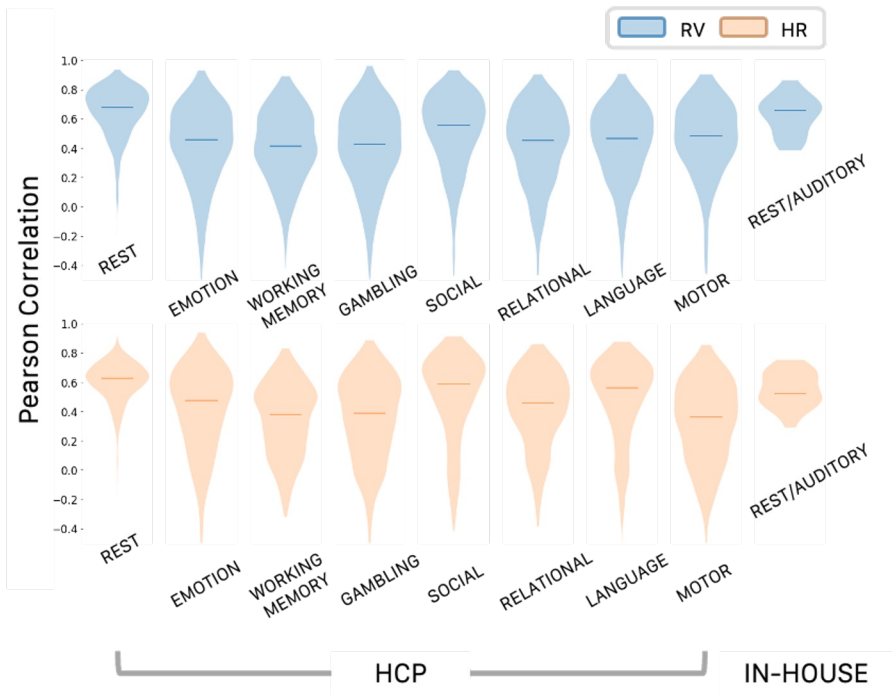


Vesselness density



Potential for denoising waveforms

Trained Models Using Resting State fMRI Data



Bayrak et al., Imaging Neuroscience doi: 10.1162/IMAG.a.163
<https://github.com/neurdylab/deep-physio-recon>

Generalization to older adults (36-89 years old)

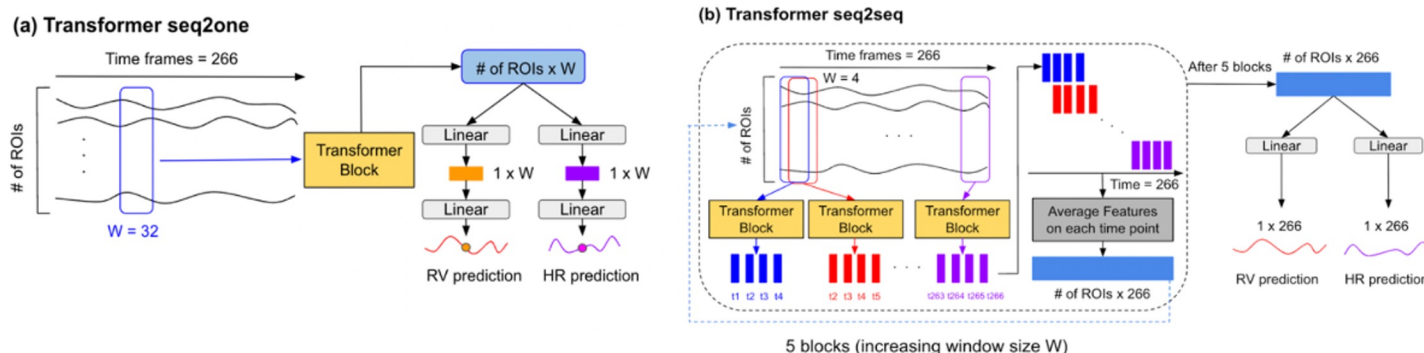
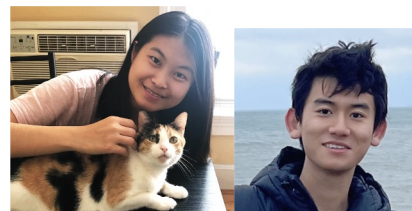


Figure 1. Transformer seq2one (a) and Transformer seq2seq (b) model architectures. These models take fMRI time-series from specific brain regions of interest (ROIs) as input, and output the predicted physiological (RV and HR) time courses.



Shiyu Wang, Ziyuan Xu et al. SPIE 2025

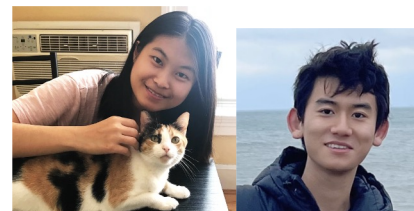
<https://arxiv.org/pdf/2408.14453>

<https://github.com/neurdylab/deep-physio-recon-lifespan>

Generalization to older adults (36-89 years old)

Table 1. RV and HR reconstruction performance on the HCP-Aging data. Values indicate Pearson's correlation between measured and recorded signals. TF: transformer

Task		RV Reconstruction				HR Reconstruction			
Model		CNN ^[17]	Bi-LSTM ^[20]	TF-seq2one	TF-seq2seq	CNN ^[17]	Bi-LSTM ^[20]	TF-seq2one	TF-seq2seq
# Parameters		0.08 M	40 M	1.5 M	22 M	0.08 M	40 M	1.5 M	22 M
Pretrain		/	0.593	/	/	/	0.553	/	/
From scratch		0.619	0.665	0.685	0.670	0.506	0.592	0.585	0.602
From scratch (jointly)		0.592	0.657	0.684	0.675	0.530	0.593	0.593	0.605
Fine-tune		0.614	0.683	<u>0.698</u>	0.684	0.525	0.606	0.595	<u>0.618</u>

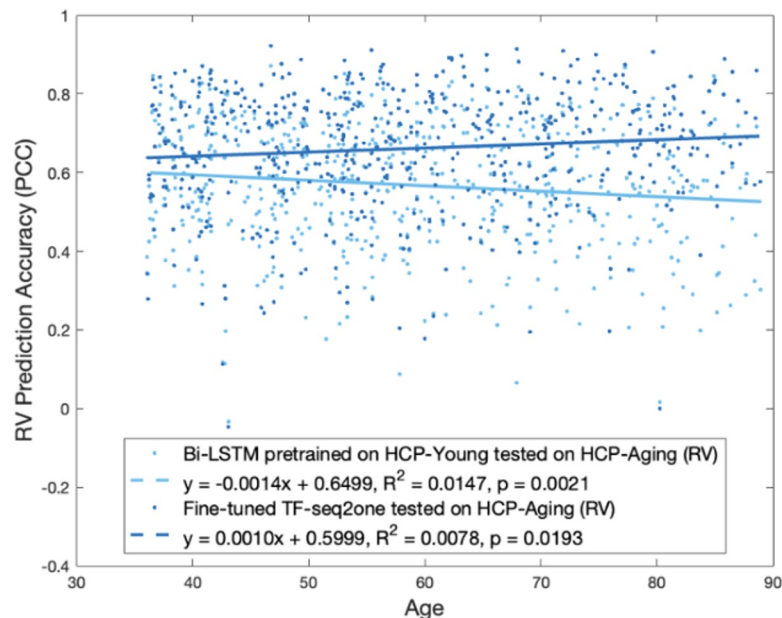
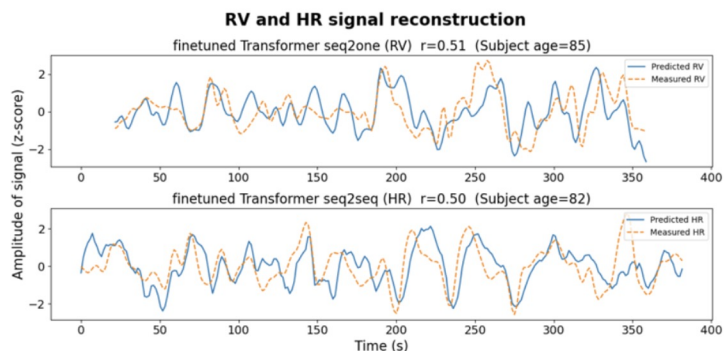
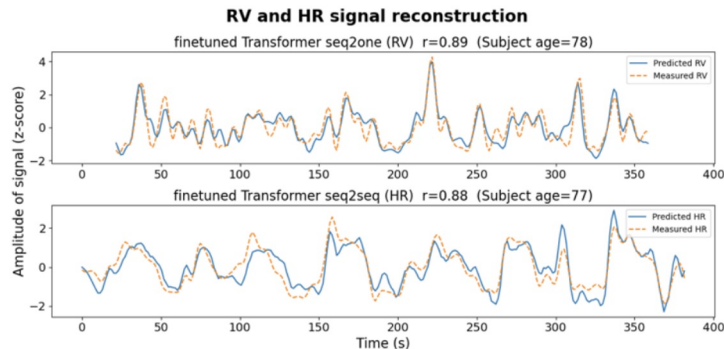


Shiyu Wang, Ziyuan Xu et al. SPIE 2025

<https://arxiv.org/pdf/2408.14453>

<https://github.com/neurdylab/deep-physio-recon-lifespan>

Generalization to older adults (36-89 years old)



1752 scans from 571 subjects median $r \sim .698$ for RV and $r \sim .618$ for HR

<https://arxiv.org/pdf/2408.14453>

<https://github.com/neurdylab/deep-physio-recon-lifespan>

Community-driven research ecosystems



 **PhysAI**

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About

Welcome to PhysAI research suite, which is dedicated to advancing the analysis of physiological signals alongside human neuroimaging data. Our tools are designed to support researchers in exploring the complex interactions between the brain and body.

Reconstruction methods enable the extraction of low-frequency cardiac and respiratory fluctuations directly from fMRI data. This is particularly valuable when physiological recordings are unavailable or compromised, enriching the analysis and interpretation of fMRI data.

Quality Assurance Tools include both manual and automated tools for the quality assessment of physiological recordings acquired during MRI sessions.

 **Software**

A suite of [tools](#) including code, models, and detailed documentation designed for the quality assurance of physiological signals and the inference of physiological dynamics from fMRI data.

 **Science**

[Research](#) initiatives focused on the complex interconnections between brain and body physiology.

ing a collaborative environment where knowledge is shared to and collaborate with open-source projects such as rts the broader scientific community.

SPIE Medical Imaging (2025), under review.

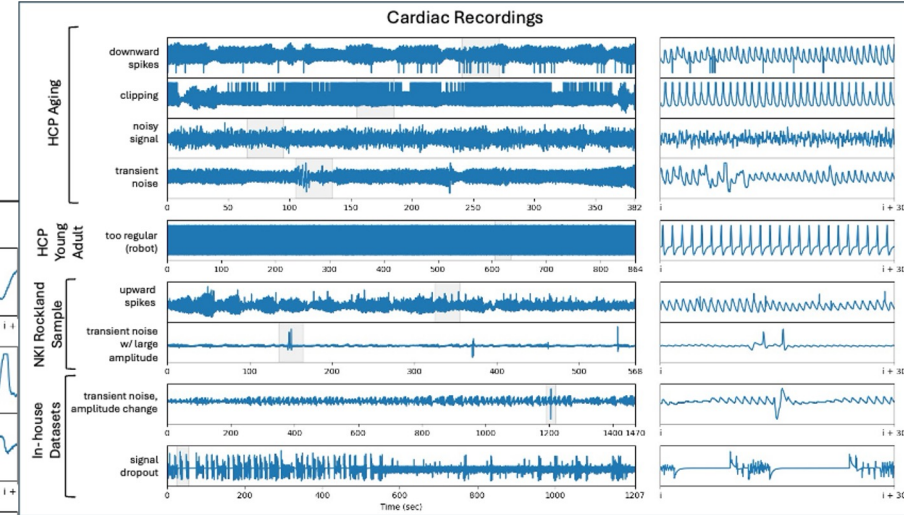
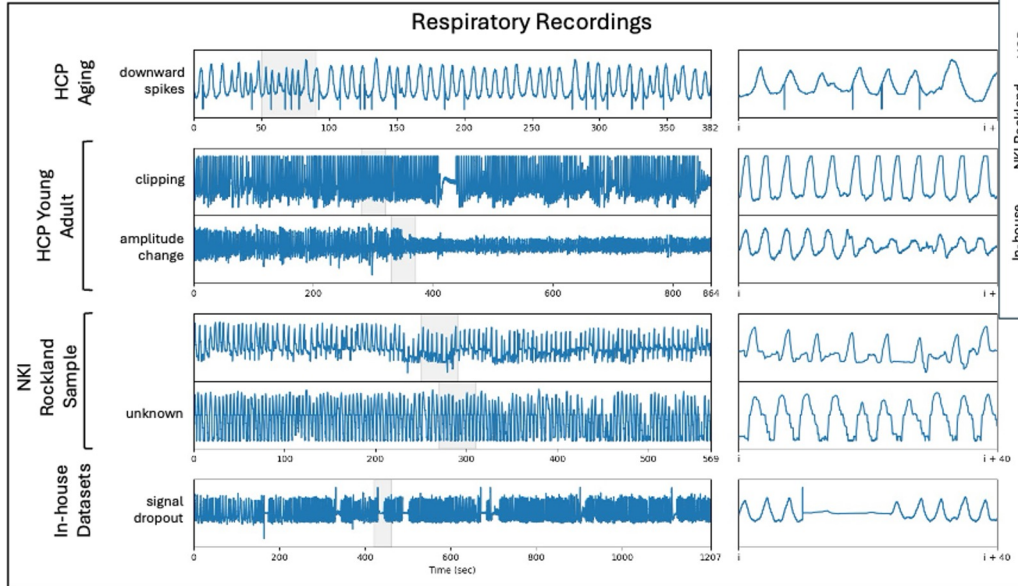
[PREPRINT <](#)

[PDF !\[\]\(18f23ec37a471f851e2e8a40dc8f10ea_img.jpg\)](#)

[CODE >](#)

<https://neurdylab.github.io/physai/>

PhysioQA - Experts



Collaborate with
us!



PhysioQA - Experts



A Collaboration to Standardize Physiological Quality Assurance in Neuroimaging

You are here because you are (we hope) interested in joining a collaborative initiative focused on understanding and documenting physiological quality assurance practices in neuroimaging.

Why this matters: As you know, there is currently no standardized approach to quality assessment across the field. As a first step, we are pooling our collective expertise and methodologies. We aim to learn, document and distribute community practices that can benefit the entire neuroimaging community.

What we're looking for: We are collecting information about how different labs approach physiological data QA, including cardiac, respiratory, CO2, eye tracking, and other measures. We are particularly interested in your specific protocols, tools, and decision-making processes.

How you can contribute: If you're interested in participating, we have created a template for you to share your lab's approach to:

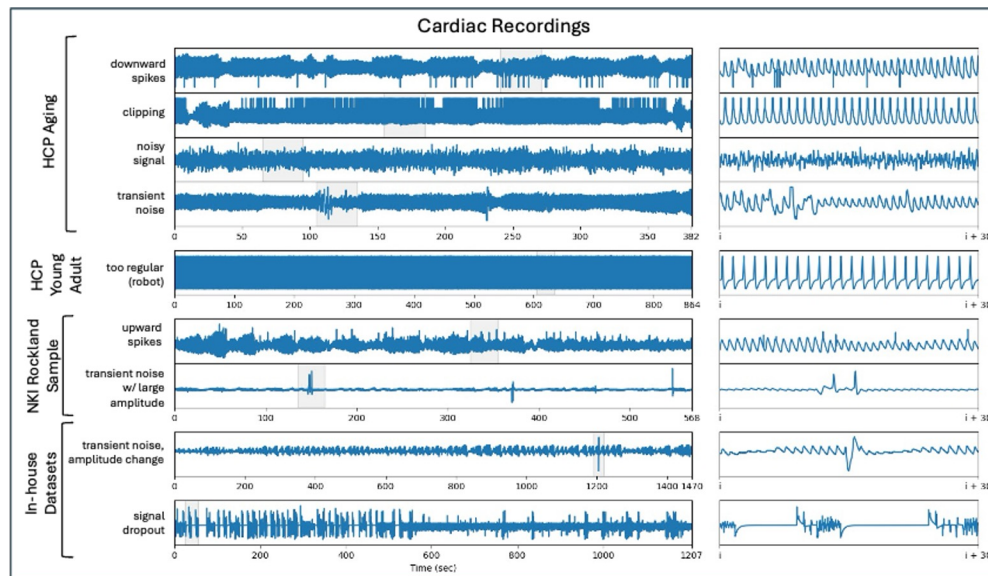
- The types of physiological data you investigate
- Your quality assessment methods and metrics
- Data correction techniques
- Datasets you've already quality-assessed
- Examples of what you consider good vs. poor quality data

Your contribution would be invaluable in helping establish community standards that improve research reproducibility and data quality.

Next steps: If you're interested in collaborating, please drop your information below, and we'll share the detailed template for you to complete. Also we're planning to compile all responses and organize a virtual meeting to discuss findings and next steps after OHBM (sometime in July, 2025)!

Thank you for considering this invitation. We look forward to potentially working together on this.

Cheers,
Roza Bayrak & Catie Chang @ NEURDY Lab



Collaborate with
us!



PhysAI/QA - Interactive Tools

Quick guide to use DataLad to download BIDS-compliant datasets

Below is a minimal, practical workflow to **install** a BIDS dataset, **retrieve** actual file content, **unlock** files of a dataset in order to access the actual content in working tree, and **undo the retrieval** of file content.

- ### 1. Prerequisites

```
# macOS (brew) examples
brew install datalad git-annex git-lfs

# Debian/Ubuntu (apt) examples
sudo apt-get update
sudo apt-get install -y datalad git-annex git-lfs
```
- ### 2. Install the dataset (symlink clone)

This command creates a local sibling (symlinks) of an existing dataset from a remote locations specified as a URL.

```
# Replace URL with the dataset source (GitHub, GIN, OSF, etc.)
datalad install -r -g -s <DATASET_URL>
# Example:
datalad install git@github.com:OpenNeuroDatasets/ds000001.git
```

The repo is usable immediately; large files are annexed and fetched on demand.
- ### 3. Retrieval of content

Get any dataset content (files/directories/subdatasets).

```
# Selective retrieval (recommended)
datalad get <PATH/TO/FILES>
# Example:
datalad get sub-*/ses-*/func/*physio*

# Get EVERYTHING (not recommended)
datalad get .
```

Got It

Physiological Signals

Reviewing: ds000210

All 135
PPGPulseCardiac 155
Respiratory 155

All (current file) 2

☐ Cardiac
 ☐ Respiratory

Auxiliary Signals

☐ Task Events
 ☐ Motion Parameters

Alignment (ONSET)

physio_onset: 0 fmr1_onset: 0 offset: 0

Selected (0)

Analyze
Export
Clear

Add Custom Signals

Add Custom Signal

sub-01_task-cuedSGT_physio.json

```
{
  "StartTime": 0,
  "SamplingFrequency": 50,
  "Columns": ["cardiac", "respiratory"]
}
```

Reviewing: ds000210

All 135
PPGPulseCardiac 155
Respiratory 155

All (current file) 2

☒ Cardiac
 ☒ Respiratory

Auxiliary Signals

☒ Task Events

✓ Auto-loaded for sub-02_task-cuedSGT_run-01: sub-02_task-cuedSGT_run-01_events.tsv

Select Event Columns:

☐ Condition ☐ CueWord ☐ IAPIdentifier ☐ ResponseTime ☐ TrialNumber

☐ VividnessRating ☐ duration

► Upload Different Events File

☐ Motion Parameters

Alignment (ONSET)

physio_onset: 0 fmr1_onset: 0 offset: 0

Selected (2)

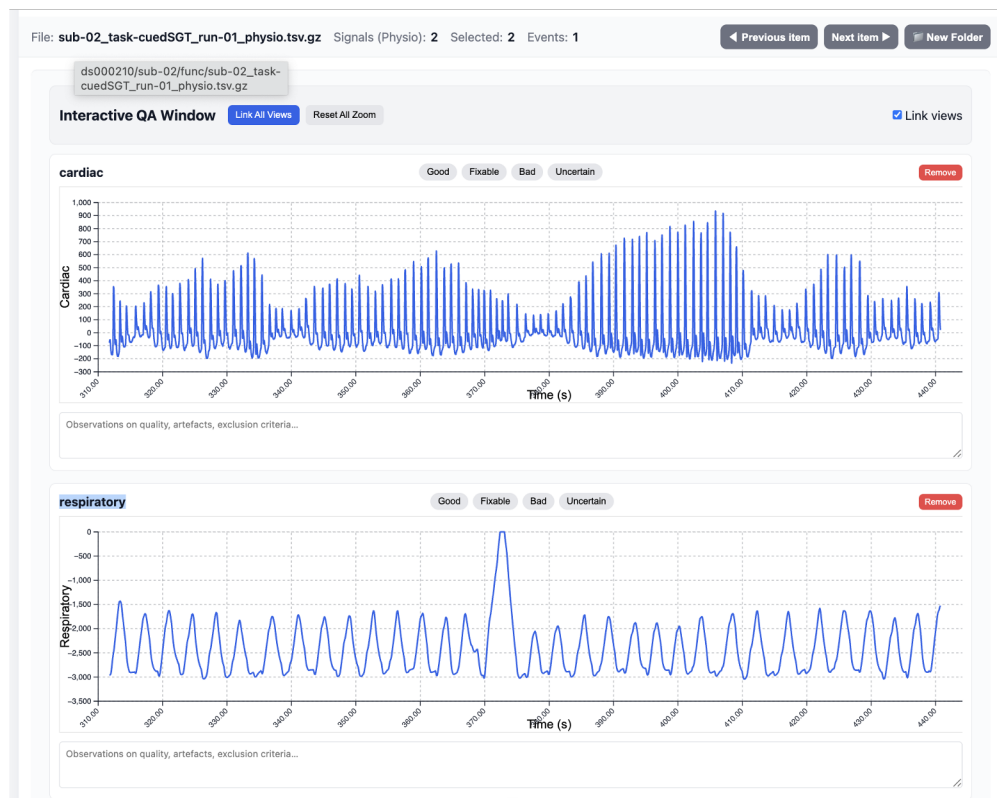
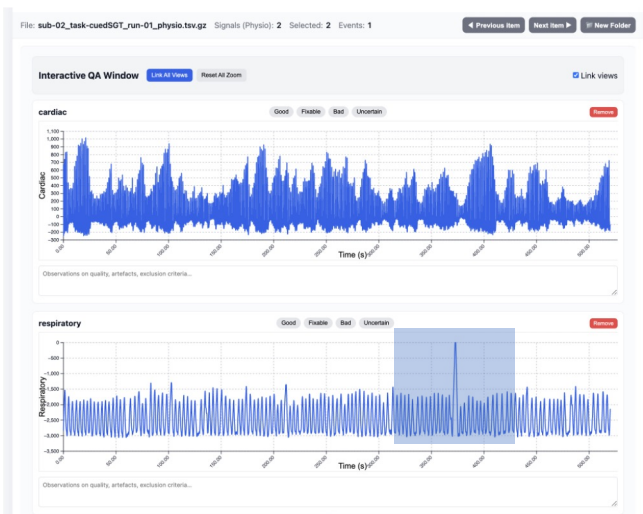
Cardiac × Respiratory ×

Analyze
Export
Clear

Add Custom Signals

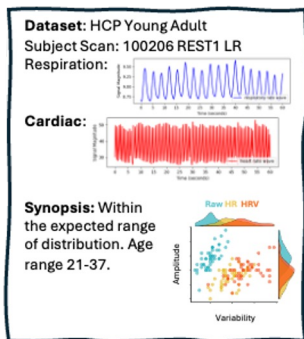
Add Custom Signal

PhysAI/QA - Interactive Tools



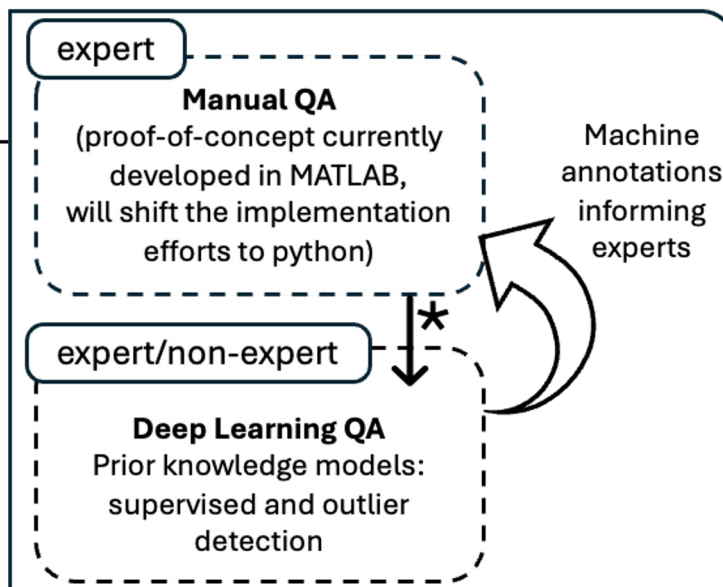
PhysAI/QA - Automated Tools

Static/Interactive
Report Generation

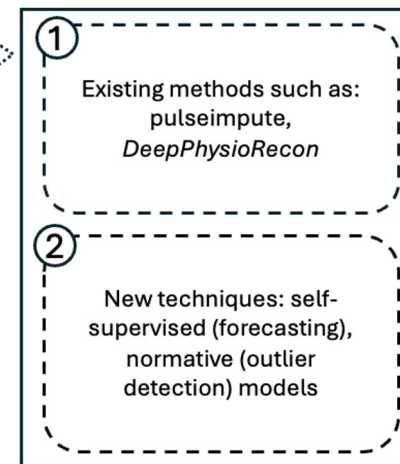


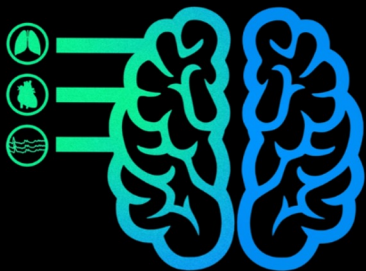
Dataset details, visualizations of raw data and derivatives, baseline QA outcome, (optional) comparative evaluation/distribution plots to other cohorts.

Quality Check



Artifact Correction





Physiopy: a Python suite for handling physiological data recorded in MRI settings

Roza G. Bayrak, Catie Chang, Katherine Bottenhorn, Molly Bright, César Caballero-Gaudes, Rebecca Clements, Inés Chavarria, Niall Duncan, Inês Esteves, Vicente Ferrer, Raphael Fournier, Daniel Glen, Sarah Goodale, Georgios Kikas, Tomas Lenc, François Lespinasse, Neville Magielse, Mary Miedema, Stefano Moia, Robert Oostenveld, Marie-Eve Picard, Joana Pinto, Céline Provins, Taylor Salo, Simon Steinkamp, Rachael Stickland, Mi Xue Tan, Hao-Ting Wang, Kristina Zvolanek, Marcel Zwiers, The Physiopy Community

Interested in collaborating? Please contact: physiopy.community@gmail.com



Contributors from our team:



Roza Bayrak



Sarah Goodale



Shiyu Wang

AIMS OF PHYSIOPY

The main goal of *physiopy* is to help collect, analyze and share physiological data. To do so by:

1. Writing packages to make **user-friendly** pipelines to work with physiological data.
2. Specializing in physiological data use in **neuroimaging** (i.e. **MRI**) data analysis.
3. Providing documentation containing **tips and strategies** on how to collect such data and use our packages.
4. Helping **set a standard** for these data, albeit without forcing users to use it.
5. (Bonus aim) It is an excuse of educational kind. We learn new topics like Git/GitHub, Python3, visualization, Physiology and related tools/software.

Importance of considering brain-body states for reproducible & interpretable fMRI:

- variability arising from these states can be quite substantial
- understanding/modeling their signatures can lead to clearer and more sensitive inferences
- recording peripheral signals can inform interpretation of fMRI and aid with modeling dynamics
- considering brain/body states can enable:
 - > advancing basic scientific understanding
 - > investigating disruption in disease

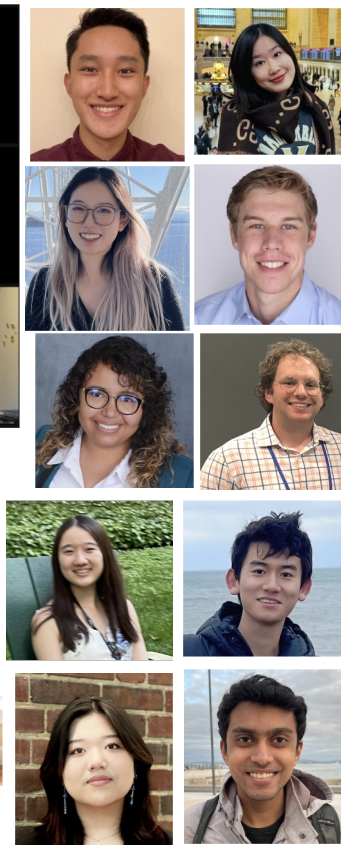
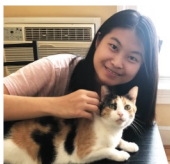
Acknowledgments

Neuroimaging & Brain Dynamics Lab ([neurdylab](#) 🧠💡) @ Vanderbilt University



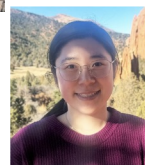
Sarah Goodale
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Roza Bayrak
Benjamin Gold
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Collaborators on this work:

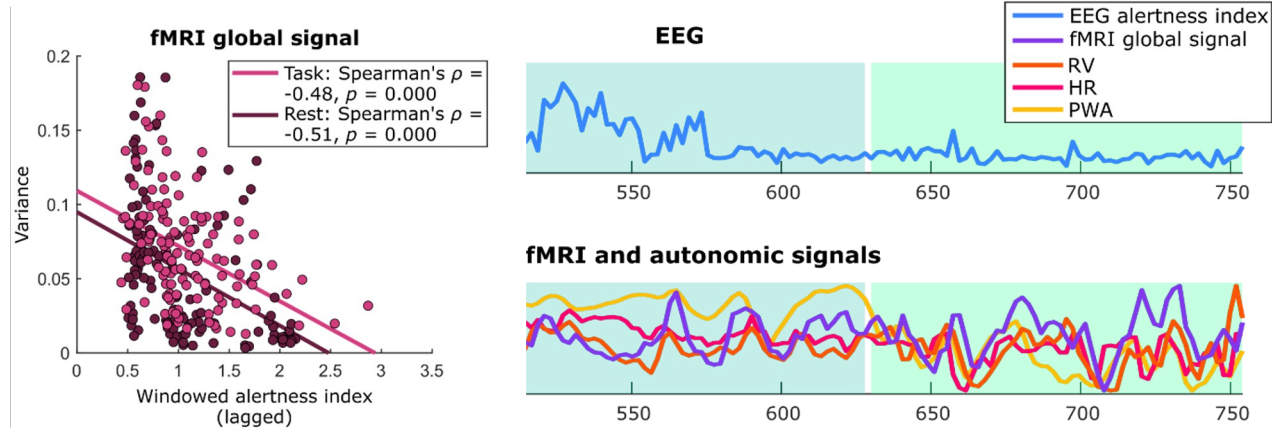
Yuankai Huo, VU
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Jeff Duyn, NIH
Taylor Bolt, UCLA
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NS112252

(extra)

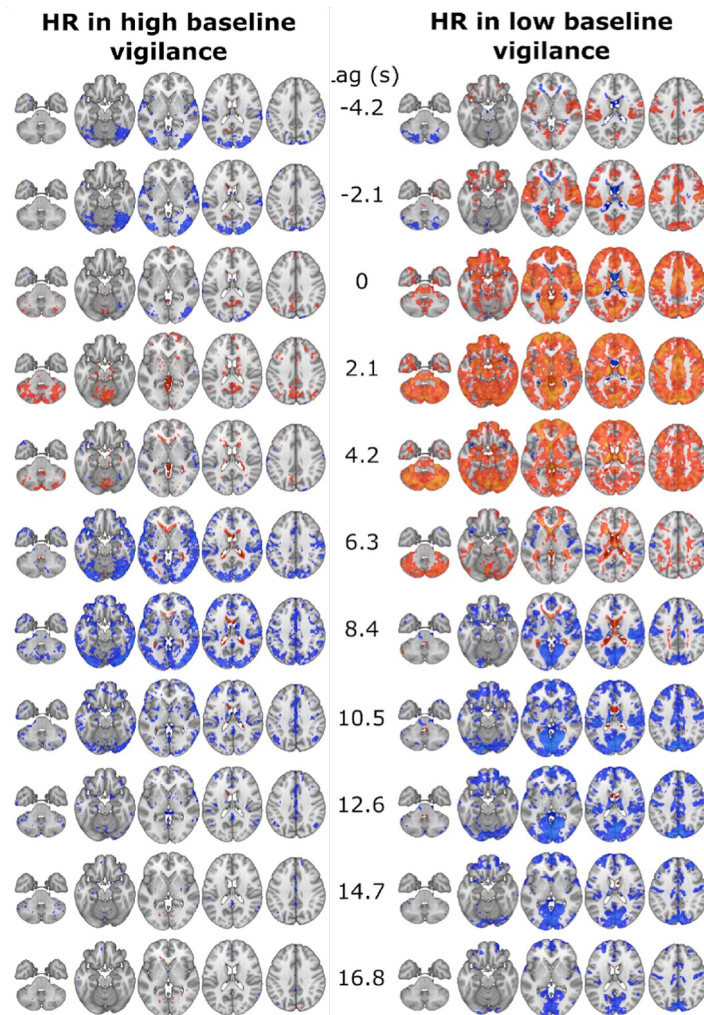
Physiological contributions to fMRI signals increase during drowsiness and light sleep



Gold et al., Imaging Neuroscience 2024

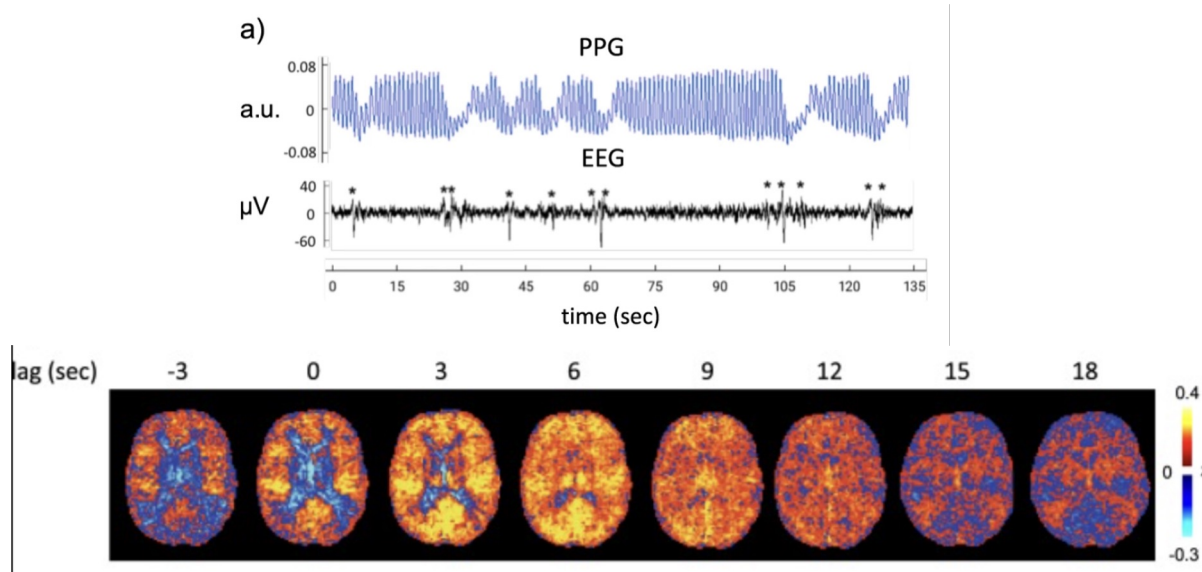
Ben Gold





Brain-body interactions: Integrating imaging and peripheral physiology

Whole-brain correlates of K-complexes (sleep)



Pinar Özbay et al., 2019